

1 Background

1.1 Problem Statement

Tiffin operates a multi-source food ordering system through its **website** and third-party platforms such as **Foodpanda, Foodie, and Pathao**, with **Chilly POS** handling order management. The current system has several operational limitations:

- Order notifications can be missed or stop after a short duration
- No alerts are received if the POS browser tab is closed
- Alerts are not persistent until an order is accepted
- No escalation mechanism for unattended orders
- Inefficient handling of orders across multiple branches

Since Chilly POS is a third-party SaaS platform, direct modification is not possible. Therefore, a **middleware-based notification system** is required to ensure reliable, real-time order alerts to branch staff.

1.2 Assessment Scope

The proposed system will address the following:

- Capture and normalize orders from multiple sources
- Deliver real-time alerts to branch devices (mobile/web)
- Ensure continuous alerting until order acceptance
- Provide a dashboard to monitor order status (pending/accepted)
- Support multi-branch operations with branch-specific routing

1.3 Required Assets from Tiffin

To proceed with development, the following assets and access will be required:

- Access to the website order flow or API (if available)
- Access to **Chilly POS** interface or notifications (for workaround integration, if permitted)
- Branch and staff configuration details (for order routing and notifications)
- Sample order data for development and testing

Third-Party Integration Requirements

- API access or partner integration approval from platforms such as Foodpanda, Pathao, and Foodie (if available)

- Required API credentials, including:
 - API keys / access tokens
 - Webhook configuration details (if supported)
 - Authentication credentials or documentation
- Access to sandbox/test environments (if provided by third-party platforms)

Additional Assets (Optional)

- Branding assets (logos, colors, notification tone preferences)
- Preferred alert rules or escalation policies

2 Proposed Service

2.1 Technical Workflow

- 1. Order Ingestion Layer**
 - Receives orders from website and third-party platforms (API, webhook)
 - Normalizes incoming data into a unified format
- 2. Core Backend System**
 - Stores order data in a database
 - Assigns orders to appropriate branches
 - Tracks order status (Pending → Accepted)
 - Publishes events for real-time processing
- 3. Real-Time Notification Layer**
 - Sends instant alerts to branch staff devices
 - Supports continuous ringing/notification until accepted
- 4. Mobile App / Web Dashboard**
 - Staff receives alerts and views order details
 - Staff can accept orders
 - Dashboard updates in real-time
- 5. Escalation System (Optional)**
 - If orders are not accepted within a defined time:
 - Notify additional staff
 - Trigger higher-priority alerts

2.2 Scope

- Order ingestion from website and third-party sources
- Real-time alert system for branch staff
- Continuous alert mechanism until order acceptance
- Multi-branch routing and management

- Dashboard for monitoring order lifecycle
- Deployment support and system setup

2.3 Out of Scope

- Direct modification of Chilly POS
- Guaranteed deep integration with third-party platforms without API access
- Advanced analytics/reporting systems
- Payment processing (handled externally)
- Server provisioning, infrastructure setup, and associated hosting costs

Note: Deployment will be handled by our team, but server procurement, configuration, and ongoing maintenance will remain the client's responsibility.

3 Implementation Plan & Timeline

3.1 Project Timeline Overview

Phase	Description	Estimated Time (Day)
Phase 0 – POC	Order flow, real-time notification, simple dashboard	5
Phase 1 – MVP	Core backend, branch logic, continuous alerts, improved dashboard	15
Phase 2 – Integration	Website integration + POS workaround + third-party APIs	20
Phase 3 – Mobile App	Dedicated mobile app with push notifications & background alerts	15
Phase 4 – Advanced Features	Escalation system, admin panel, monitoring & logs	20
	Total	75

3.2 Project System Diagram

